

Algebra Basics

Introduction to Algebra

Algebra is a branch of Mathematics that uses:

- Numbers
- Symbols
- Variables (letters)

Algebra helps us represent unknown values and solve problems easily.

Example

$$x + 5 = 12$$

Here:

- $x \rightarrow$ unknown value
- $5 \rightarrow$ constant value
- $12 \rightarrow$ total value

After solving:

$$x = 7$$

Why Do We Learn Algebra?

Algebra helps us:

- Solve unknown values
 - Understand formulas
 - Build logical thinking
 - Solve real-life problems
 - Learn higher Mathematics
 - Understand Physics and Programming concepts
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Important Algebra Terms

Term	Meaning	Example
Variable	Unknown or changing value	x, y
Constant	Fixed value	5, 10

Term	Meaning	Example
Coefficient	Number attached to variable	3 in $3x$
Expression	Combination without = sign	$2x + 5$
Equation	Mathematical statement with =	$x + 2 = 5$
Like Terms	Same variables and powers	$2x, 5x$
Unlike Terms	Different variables or powers	$2x, 3y$

1. Variables

Definition

A variable is a letter used to represent an unknown value.

Common Variables

- x
- y
- a
- b

Example

$$x + 3 = 7$$

Here x is the variable.

2. Constants

Definition

A constant is a fixed value that does not change.

Examples

- 2
- 5
- 100

Example Expression

$$x + 5$$

Here 5 is constant.

3. Coefficient

Definition

The number multiplied with a variable is called coefficient.

Example

$5x$

Here:

- $5 \rightarrow$ coefficient
- $x \rightarrow$ variable

4. Algebraic Expressions

Definition

An algebraic expression is a combination of variables, constants, and operations.

Examples

- $x + 2$
- $3a - 5$
- $x^2 + 3x + 1$

Important Note

Expressions do NOT contain = sign.

5. Terms

Definition

Parts of an expression separated by + or - are called terms.

Example

$2x + 5 - 3y$

Terms are:

- $2x$
 - 5
 - $-3y$
-

6. Like Terms

Definition

Terms having same variables and same powers.

Examples

- $2x$
- $5x$
- $-3x$

Simplification

$$2x + 5x = 7x$$

7. Unlike Terms

Definition

Terms having different variables or different powers.

Examples

- $2x$
- $3y$
- x^2

These cannot be added directly.

8. Equations

Definition

An equation is a mathematical statement with an equal sign.

Example

$$x + 2 = 5$$

9. Exponents (Powers)

Definition

Exponent tells how many times a number is multiplied by itself.

Example

$$2^3 = 2 \times 2 \times 2 = 8$$

Parts

- 2 → base
 - 3 → exponent
-

10. Polynomials

Definition

An expression containing variables with powers.

Examples

- $x + 2$
 - $x^2 + 3x + 1$
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Types of Polynomials

Type	Meaning	Example
Monomial	One term	$5x$
Binomial	Two terms	$x + 2$
Trinomial	Three terms	$x^2 + x + 1$

11. Degree of Polynomial

Definition

Highest power of variable in polynomial.

Example

$$x^3 + 2x^2 + x$$

Highest power = 3

Degree = 3

12. Simplification

Definition

Combining like terms to make expression easier.

Example

$$2x + 3x = 5x$$

13. Algebraic Identities

Important Formulas

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$a^2 - b^2 = (a - b)(a + b)$$

14. Factorization

Definition

Breaking expression into smaller factors.

Example

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

Real-Life Uses of Algebra

Shopping

If one pen costs ₹10:

$$10x$$

Here x = number of pens.

Age Problems

Mother age = x

Father age = $x + 5$

Distance Formula

Distance = Speed $\times$ Time

Common Mistakes

Wrong Addition

$$2x + 3y \neq 5xy$$

Wrong Exponent Rule

$$x^2 + x^2 = 2x^2$$

NOT x^4

Practice Questions

Easy Level

1. Simplify:

$$2x + 5x$$

1. Simplify:

$$3a - 2a$$

1. Solve:

$$x + 5 = 11$$

1. Find:

$$2^4$$

1. Find degree:

$$x^3 + 2x + 1$$

Summary Table

Topic	Main Idea
Variable	Unknown value
Constant	Fixed value
Expression	No = sign
Equation	Contains = sign
Polynomial	Expression with powers
Exponent	Repeated multiplication
Simplification	Combine like terms

Next Topics to Learn

1. Operations on Expressions
2. Linear Equations
3. Polynomials
4. Factorization
5. Coordinate Geometry

6. Quadratic Expressions

7. Algebraic Identities